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Introduction

The passenger/crew door is the primary entrance and exit for the aircraft. The door gives also support to the cabin and aerodynamic pressure during flight.

05/03/21

General description

The aircraft has a semi-plug type passenger/crew door installed on the left side of the forward fuselage. The door gives access to the passenger and flight compartment and is also used as an emergency exit for evacuation.

The passenger/crew door includes the components that follow:

- The passenger/crew door structure
- The passenger/crew door actuator and cable
- The passenger/crew door seal
- The passenger/crew door side-covers
- The passenger/crew door handles
- The passenger/crew door handrails
- The airstairs and folding steps
- The passenger/crew door gas-springs
- The passenger/crew door telescopic-struts
- The vent flap and vent flap seal
- The passenger/crew door switches
- The passenger/crew door mechanisms
- **The passenger/crew door handrail extension.**

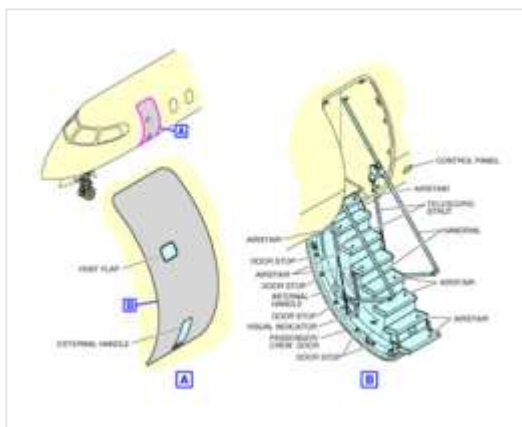


Fig 1 Passenger/crew door - Location

When the door is unlatched, unlocked, and lifted, it rotates out and down by gravity and stops immediately above ground level. The door rotates around the bottom hinges or door arm during the opening and closing. The opening motion of the door is dampen by the door actuator located in the under-floor structure and by the telescopic struts installed along the airstairs. The door can be opened from the inside or outside by handles (internal and external) and closed by the actuator. The actuator is commanded by switches installed on the passenger/crew door control panel. There is one control panel installed outside on the fuselage at the aft of the door and one in the cabin. The door can also be closed manually without electrical power thanks to a handle installed on the actuator. A viewing aid makes sure that the area around the door is clear before the opening.

The airstairs installed in the door structure gives immediate egress to the aircraft. The handrails installed along the airstairs extend and retract during the opening and closing of the door. The door is installed with door stops and guide fittings that transmit pressurization loads to the aircraft structure.

A red/green visual indicator on the door shows the door closed, latched and locked status. The door has proximity switches to sense the latched, locked, closed and incorrect status of the door. These switches are connected to the Landing Gear and Steering Control Unit (LGSCU). The LGSCU sends the data to the Data concentrator Module Cabinets (DMCs) that process the data in the internal logic for Engine Indication and Crew Alert System (EICAS) and Onboard Maintenance System (OMS) display.

The passenger/crew door has interfaces with the systems that follow:

- Cabin pressurization system (ATA 21-31-00)
- Electrical system (ATA 24-00-00)
- Landing-gear position-indication system (ATA 32-61-00)
- Lighting system (ATA 33-00-00).

Passenger/crew door structure

The passenger/crew door is installed on the left side of the fuselage between FS320.00 and FS357.00. The door structure transmits the pressurization loads to the aircraft structure through the door stops.

The door structure consists of:

- The door skin
- The beams
- The stabilizing rods
- The frames
- The guide fittings
- The door stops and pins
- The splices
- The handle box
- The door arm (bottom hinge).

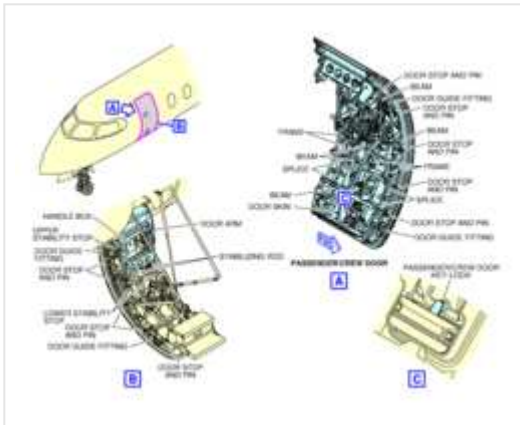


Fig 2 Passenger/crew door structure - Location

The passenger/crew door skin is a machined surface made of aluminum alloy sheet metal. It includes the provision for the vent flap and the external handle housing. The frames and beams are machined parts made of aluminum alloy. These frames and beams transmit loads to the door stops which are then transmitted to the fuselage. The splices are made of aluminum alloy sheet metal and are assembled between the frames and beams to make the load transfer easier. The door has sufficient drainage with the aid of the dry liner between the door skin and the insulation to protect against humidity and prevents water entrapment.

The passenger/crew door includes a key lock installed on the external handle. The lock keeps the door in the locked position when closed.

The door structure is installed on a door arm (bottom hinge) attached to the fuselage. The door arm holds the weight of the door during its operation and receive the loads transmitted from the door. The arm is a mechanical assembly of machined parts made of aluminum alloy with a sheet metal skin. To keep the door stable in an open configuration, two stabilizing rods are installed in opposite directions to limit the door structure displacement.

The four guide fittings are assembled in relation to the lift curve. Two guide fittings are at the lower side of the door and two are at the top of the door. The counter parts of the guide fittings are the four fuselage guide rollers.

Ten door stops (five on each side) are installed on the door stop fittings to transmit the pressurization loads to the fuselage. When the door is closed, they touch the fuselage stop pads which receive the loads and transmit them into the aircraft structure. The door stop pins are mechanical assemblies that are designed to receive high loads and supply low friction to the fuselage stop pads. The tip of the pin is a bearing type material that has a misalignment of 3 degrees in all directions between the stop pin and the fuselage stop pad. An O-ring seal safeties the assembly.

The door follows a trajectory given by the door guide fittings when opening. This trajectory is made to keep the door stops clear of the fuselage stop pads. The trajectory is 0.12 in. (0,30 cm) inward movement of the door in relation to the Mean Pressure Plane (MPP). This trajectory is approximately 11 degrees to the MPP and lets the door go out of the door stop with a minimal sliding to decrease wear.

For stability, two stability stops on the lower side and two stability stops on the upper side of the door arm are included in the door arm linkage. When the door is fully opened, the stability stops keep the position of the door stable in relation to the door arm. The lower stability stops are controlled directly by the handles. The stability stops are adjustable and will be correctly positioned once the door is adjusted on the aircraft.

Passenger/crew door actuator and cable

The passenger/crew door actuator is installed in the under-floor at FS339L on the forward side of the door. The door actuator is used to lift the door with cables when closing and it is supplied by 28V DC electrical power.

The functions of the door actuator are to:

- Close the door electrically by speed reduction until the door is in its closed position.
- Give a variable damping speed during door opening without an external power supply.
- Close the door manually if the door actuator motor is defective.

The door actuator includes the components that follow:

- The cable and drum
- The gear reducers
- The Electric Motor and Control Module (EMCM)
- The manual closing-device handle.

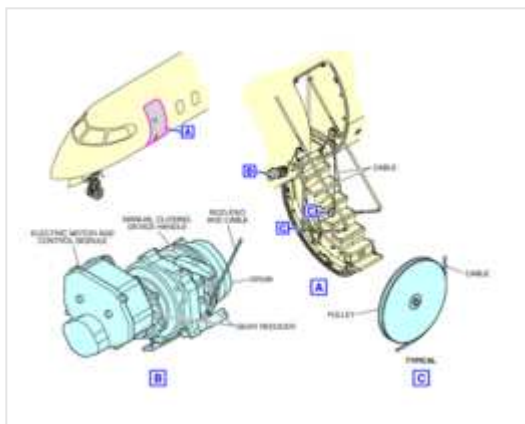


Fig 3 Passenger/crew door actuator and cable - Location

The door actuator is commanded by one internal and one external door switch. The length of the cable can be adjusted by rod ends to have an equal tension on the aft and forward sides.

The design of the door actuator and its electronic components gives better efficiency for the closing time in all temperatures range. During the closing of the door, the control unit will decrease the door speed when the door is closer to the fuselage to prevent damage to the door. The actuator enters in damping mode 10 seconds after the release of the door switch.

The door actuator does not receive power until the door has completed the lift motion. Once the door is lifted, the actuator is powered by a micro-switch. If the door opening has started, it has to get to its complete opened position before the actuator can be commanded again. If there is an electrical failure, the door can be closed from the inside with the manual closing-device handle attached to the actuator.

Passenger/crew door seal

The door has a seal made of silicon rubber which is reinforced with fabric. The seal makes the contact with a permanent fixed striker-seal (depressor) around the periphery of the door surround to make a pressure barrier when the door is closed. The seal is divided into segments to prevent propagation of seal tear (if there is external damage on the seal). The seal always keeps a positive contact with the striker seal in relation to the maximum door rigging tolerances, maximum displacement of the door and with aerodynamic loads. The seal also gives water sealing if there is a ditching event.

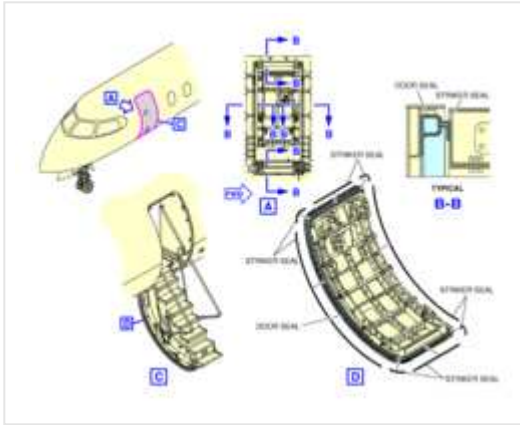


Fig 4 Passenger/crew door seal - Location

Passenger/crew door side-covers

Two side covers are attached to the sides of the passenger/crew door to protect against damage.

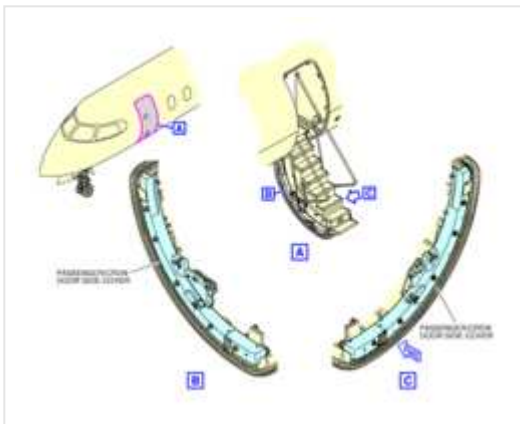


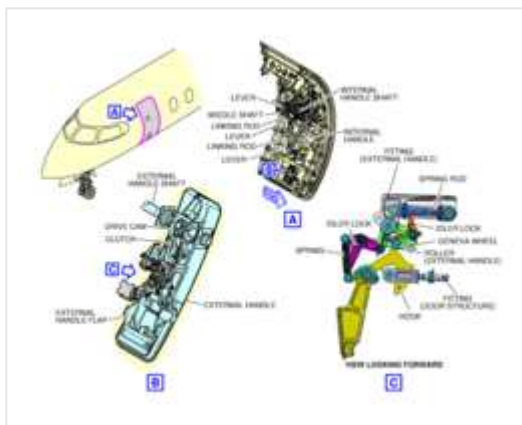
Fig 5 Passenger/crew door side covers - Location

Passenger/crew door handles

The passenger/crew door has an internal handle and external handle. The external handle is found on the lower and central zone of the door. The internal handle is found on the forward side of the door. The rotation angle is 120 degrees for the external handle and 100 degrees for the internal handle.

The handles are connected through a handle linkage mechanism that includes:

- The internal handle shaft
- The external handle shaft
- The middle shaft
- The linking rods
- The levers.

**Fig 6 Passenger/crew door handles - Location**

The internal handle commands most of the mechanisms such as the vent flap mechanism, locking mechanism, latching mechanism, and lifting mechanism.

The middle shaft which is connected at the end of the internal handle shaft commands the interlock and the stabilizing mechanism of the door.

The external handle moves the external handle shaft with a clutch installed on the handle and connects with the drive cam installed on the external handle shaft. The clutch separates the internal and external handles. If the internal handle is used for the opening and

closing, it will have no effect on the external handle position. If the external handle is used, the position of the internal handle is not important.

The external handle has a handle flap on the bottom to hold and operate the handle. The handle flap is connected to a hook and springs that keep the external handle in the stowed position during flight. To stop the play of the handle in the hooked position, a geneva wheel mechanism is included. In the geneva wheel mechanism, an over-centered spring rod keeps the external handle stowed in its housing. If the external handle is in the closed or open position, the geneva wheel mechanism will stay in the open configuration. If the geneva wheel mechanism is closed while the external handle is in the open position then if the operator tries to close the door, the external handle roller mechanism will make contact with the outer surface of the geneva wheel mechanism and it will not be possible to close the door.

An idler locks the handle flap in the open position to prevent injury to the operator while opening or closing the door with the external handle. The position of the idler is controlled by a spring and by the geneva wheel mechanism.

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Passenger/crew door handrails

The handrails system installed on the passenger/crew door is used for the safety of the passengers and crew during entry and exit of the aircraft. The handrails will be deployed immediately when the door starts to open.

The handrail system has three sets of bars installed symmetrically on each side of the door, which include:

- Two telescopic handrails, which are attached to the interior cabinets at one end.
- Two main handrails, which are used to step up or step down the airstairs.
- Two handrail posts, which are structural elements designed to support the handrails.

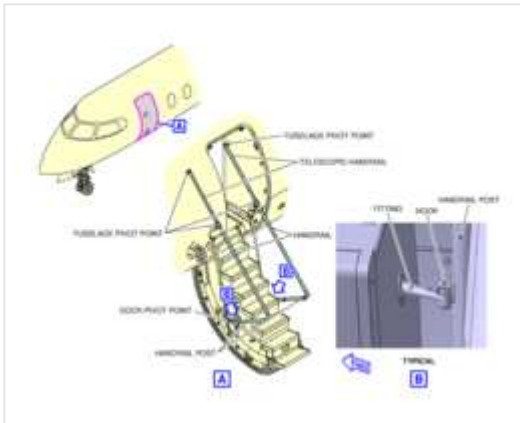


Fig 7 Passenger/crew door handrails - Location

The handrails supports are installed on the two sides of the airstairs. They have four hooks installed on the handrail post and four fittings installed on the two sides of the airstairs. Two fittings are installed on two hooks to prevent the vibration of the handrails or noise after the door is in the fully closed position. When the door is fully opened, the telescopic handrails and the handrails will align in a straight line. The fittings will fit into the hooks to stiffen the handrails after the door is fully opened. **On aircraft post SB700-52-7501, handrail extension is installed on the passenger/crew door.**

Airstairs and folding steps

Airstairs

The airstairs receive the loads from the passengers weight. They are made of a single composite part which is attached directly to the door structure with quick release pins to make the maintenance and inspection operations easier. The storage bins are installed at the bottom of the third airstair are on the left and right sides of the airstair assembly.

Folding steps

The folding steps are included at the top and at the bottom of the door. They retract and extend in relation to the door position. The step trim panel is attached to the floor at the bottom part of the door.

There is one upper folding step and two lower folding steps. When the door is open, the door arm found at the bottom of the door, attaches to the lower folding step and makes one single step. The command of the lower folding steps comes directly from the pivot shaft of the handrails at the middle of the door. In relation to the position of the handrails, the two folding steps are deployed or retracted, the timing is done by the mechanism linkage. The command is received only by one lower folding step which transmits it to the second through two plate rods.

The upper folding step is operated by the door arm. A rod is attached to the door arm and connected to the upper folding step link which has a roller in contact with the upper folding step. When the door opens, the link holds the upper folding step to make sure that there is a smooth deployment. When closing the door, the link pushes the upper folding step through its roller to the closed position.

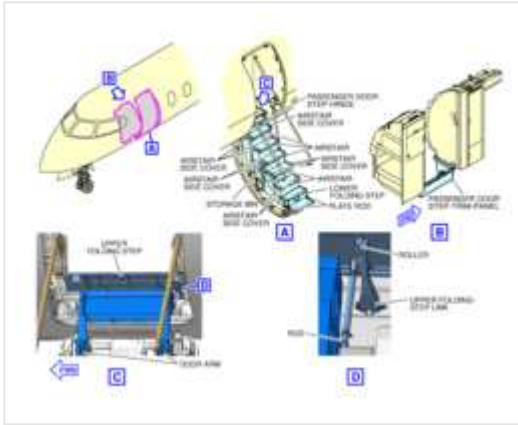


Fig 8 Airstairs and folding steps - Location

Passenger/crew door gas-springs

Two gas springs are attached to the forward and aft sides of the door between the door arm and the structure. They are part of the weight compensation mechanism. Their function is to balance the weight, to decrease the handle loads and make the door manipulation easier. The gas springs tend to lift the door during the full sequence of the opening or closing of the door.

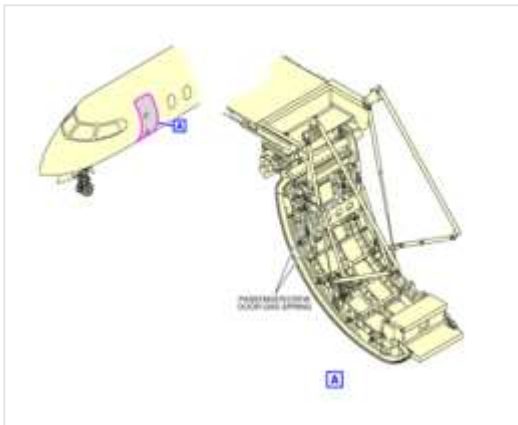


Fig 9 Passenger/crew door gas-springs - Location

Passenger/crew door telescopic-struts

Two telescopic struts are installed along the airstairs. They are used to dampen the door during the opening to prevent the mechanical shocks that can damage the door. The door actuator is used to dampen the door if the telescopic strut fail.

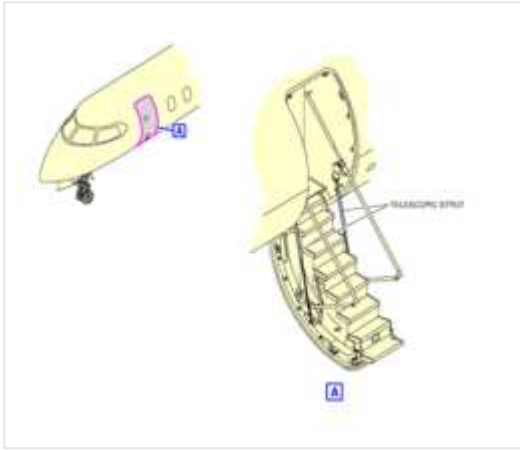


Fig 10 Passenger/crew door telescopic-struts - Location

Vent flap and vent flap seal

The vent flap is installed with a vent flap seal on the passenger/crew door cutout on the external skin. The vent flap geometric area is 50 in² (322,58 cm²). The vent flap releases the residual pressure of the cabin during the opening of the door. It is operated by the internal handle and the external handle through the mechanical linkages of the vent flap mechanism.

The vent flap is designed to make sure that, in normal operation, it cannot be closed until the door is fully closed, latched, and locked. The vent flap area prevents the initiation of the pressurization at maximum Environmental Control System (ECS) flow rates with the two outflow valves closed at a maximum cabin differential pressure of 0.5 psi (3,44 kPa). The vent flap will not prevent the opening of the door if there is a maximum differential pressure is less than 0.46 psi (3,17 kPa). The vent flap is connected with a pressure lock device to prevent its opening during flight.

The vent flap seal is installed on the circumference of the vent flap to make a pressure barrier when the door is in the closed position. The counterpart of the vent flap seal is the skin doubler installed on the door skin.

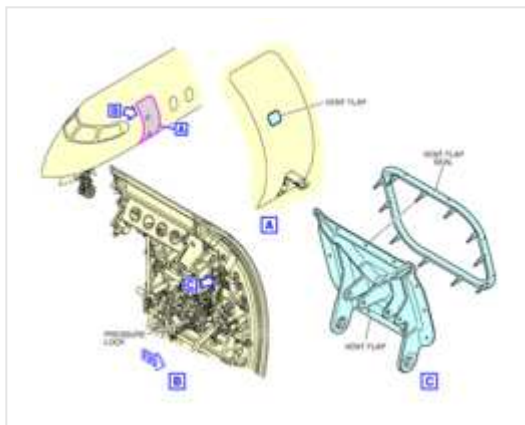


Fig 11 Vent flap and vent flap seal - Location

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Passenger/crew door switches

The passenger/crew door can be closed from the cabin with the internal switch and from outside with the external switch in the door control panel. The external switch is installed at the aft part of the passenger/crew door on the fuselage structure between FS361.27 and FS372.00 behind an access panel. The internal switch is installed on the **entrance enclosure** at FS305L adjacent at the top **wardrobe** door.

These switches are two-position toggle switches that operate the door actuator to close the door with electrical power. The center position of the switch is held and keeps the current state of the door. The up position is kept momentarily. In the up position, the switch toggles and is held in the up position during the closing movement of the passenger/crew door.

A passenger/crew door internal-switch is installed on the passenger/crew door and prevents the supply of power to one of the two closed internal and external switches when the door is closed, latched and locked.

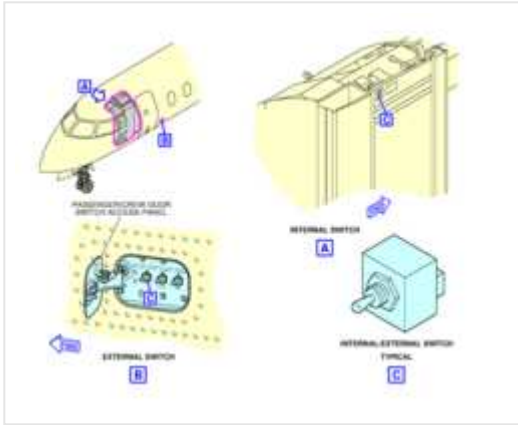


Fig 12 Passenger/crew door switches - Location

Passenger/crew door mechanisms

The passenger/crew door has mechanisms to assist during the operation and to secure the door during flight.

The door mechanisms consist of:

- The latching mechanism
- The locking mechanism
- The vent flap mechanism
- The pressure lock mechanism
- The lift mechanism
- The weight compensation
- The hold-up mechanism
- The hold-open mechanism
- The interlock mechanism.

Latching system

The function of the latching mechanism is to secure the door by holding it in the closed and locked position.

The latching mechanism includes:

- Latches (forward and aft)
- Cranks
- A latch rod
- A latch shaft
- A latch hook
- A compression spring.

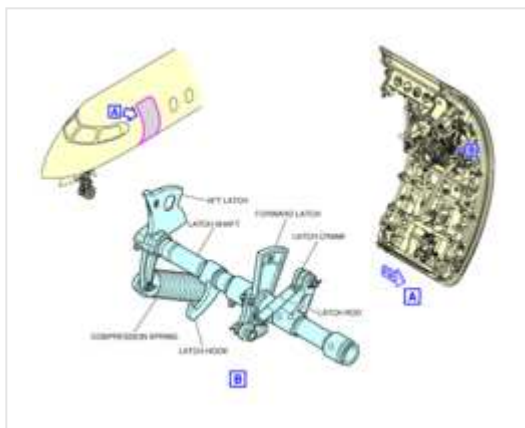


Fig 13 Latching system - Location

The passenger/crew door is latched when the latches (forward and aft) are fully engaged with their counterparts and held in position. These latches will react to loads during flight.

The latching mechanism is operated by the latch rod connected to the internal and external handles. The rod commands the latching and unlatching operations when the handles are operated. The two latch cranks are the movable elements which make the latch system a fail-safe system and prevent the door from opening. These cranks will support some loads in some flight conditions, when the airplane is unpressurized. The latch cranks are designed to make sure that all loads will be absorbed in the latch shaft bearings with no rotational movement of the latches and will not cause unlatching forces or moments.

The latch hook is installed on the latch shaft to prevent the lifting of the door, if the door is latched and the latch rod is defective. This prevents excessive loads to be transmitted in the linkage thanks to the mechanical gain of the lift cam.

The compression spring installed on the latch shaft at the aft latch keeps minimum one of the two latches engaged. The spring also gives positive means to make sure the door does not unlatch during flight because of vibrations, structural loads and deflections, positive and negative pressure loads, positive and negative “g” loads and aerodynamic loads. The status of the latching system is monitored by a proximity switch found on the aft latch.

Locking system

The function of the lock mechanism is to lock the latches and prevent the latches from unlatching. The locking system has a fail-safe design in which, if a single failure in the latching mechanism occurs, a minimum of one latch will stay locked and will prevent the accidental opening of the door.

The lock operating mechanism has:

- Locks (primary and secondary locks)
- A command rod
- A lock shaft
- A door visual indication
- A lock spring.

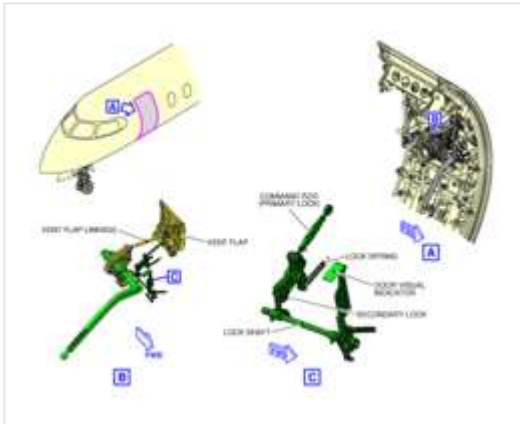


Fig 14 Locking system - Location

The locks are mechanical elements that prevent the disengagement of the latches. The primary lock locks the latch shaft and the secondary lock locks the forward latch. The forward latch has a back zone which comes in contact with the primary lock. In principle, it is not possible to unlatch the door in the locked position. The rods for the primary and secondary locks are adjustable to accurately position the locks in relation to the forward latch.

The locks are operated by the vent flap linkage. The door has one lock operated by the vent flap assembly through an adjustable lock-command rod. The vent flap controls the position of the lock and the lock controls the latch mechanism. The lock is made of a lock pawl lever that is operated from the vent door linkage and will engage with the counter-lock. This will make sure that the latches will not unlatch and the door does not open. The vent flap will not close or lock if the latch shaft is not in the latched position.

The lock is monitored by a proximity sensor in the door structure and is also mechanically shown by the visual indicator.

The lock mechanism is installed with a lock spring. If there is a failure of the lock mechanism, the lock spring will keep the door in an operational state. This will be shown on the door visual indication since the flag will stay red on the door closure.

Vent flap mechanism

The vent flap mechanism is made of an operating mechanism and the vent flap. The primary functions of the vent flap mechanism are to:

- Prevent the initiation of pressurization to an unsafe level if the door is not properly closed, latched and locked.
- Release the residual pressure during an emergency egress.
- Prevent intentional opening of the door during flight with the aid of the pressure lock mechanism.

The vent flap mechanism includes:

- A vent flap rod
- A vent flap cam
- Vent flap cam-rollers
- A double yoke
- A command shaft
- Springs.

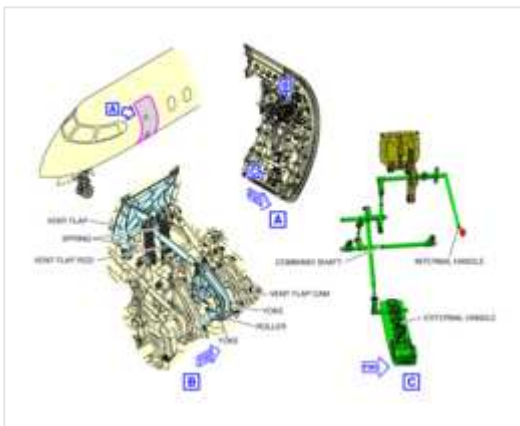


Fig 15 Vent flap mechanism - Location

The vent flap mechanism is operated by the internal handle and the external handle through mechanical linkages. When the internal or external handles rotate, they command the command shaft on which the vent flap cam is installed. The vent flap cam is an open type cam with two rollers attached to the vent flap yokes which commands the opening (roller No. 2) or the closing (roller No. 1) of the vent flap through the vent flap rod. The rollers (roller No. 1 and roller No. 2) are installed in a yoke which serves as a link between the vent flap rod and the vent flap cam. The vent flap cam in contact with the roller will give the kinematics of the vent flap.

The kinematics of the latch, lock and vent flap mechanism is adjusted by the cam profiles on the handle shaft. The linkage monitors the position of the lock. The vent flap linkage is designed to make sure that, in normal operation or with a single failure in the operating linkage, the vent flap cannot be closed until the door is latched and locked.

The two springs connected to the vent flap keep the vent flap open (if cabin pressure is less than the pressure lock threshold (0.46 psi (3,17 kPa)) to prevent the pressurization of the cabin if the door is not properly closed, latched and locked. The design of the vent flap mechanism lets the opening of the door if there is an emergency egress combined with the residual pressure of 0.125 psi (0,86 kPa).

Pressure lock mechanism

The latch/lock system is also installed with a pressure lock mechanism related to the vent flap system. The pressure lock mechanism is designed to prevent the vent flap from opening if the cabin pressure difference is more than 0.46 psi (3,17 kPa). It is designed to engage and make sure that the handle shaft does not rotate, thus preventing the unlatching and unlocking of the door.

The pressure lock mechanism includes:

- The pressure lock rod
- The pressure lock spring
- The pressure lock fitting
- The pressure lock lever.

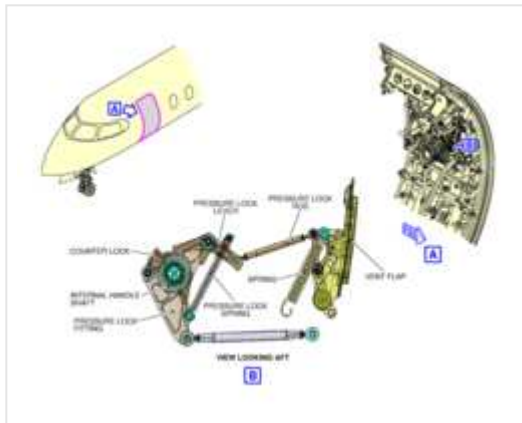


Fig 16 Pressure lock mechanism - Location

The pressure lock mechanism is made of the pressure lock rod attached to the vent flap which operates the pressure lock lever. In the closed position, the pressure lock lever causes a blockage of the counter lock installed on the internal handle shaft. If the cabin pressure difference is more than 0.46 psi (3,17 kPa), the pressure lock will prevent the rotation of the internal handle shaft, thus it is not possible to open the door. When the pressure difference has decreased to less than 0.46 psi (3,17 kPa) then the springs can open the vent flap, the pressure lock lever will disengage to stay clear of the counter-lock and the door can be unlocked, unlatched and opened.

Lift mechanism

The function of the lift mechanism is to lift the door after the unlatching to stay clear of the door stops and fuselage stop fittings.

The lift mechanism includes:

- A lift rod
- Lift cranks (forward, mid and aft)
- A lift cam
- A lift shaft
- Compression rods
- Rollers
- A lift yoke.

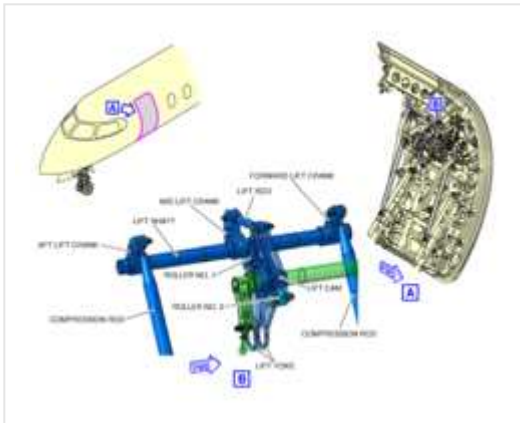


Fig 17 Lift mechanism - Location

The door lift is a direct effect from the internal or external handles which commands the lift cam. This movement is transmitted to a pull type rod attached to the lift shaft. The lift cam is of an opened cam type and commands the mid lift crank through two rollers attached to

the lift yoke that commands the lift rod.

In the sequence, the command is transmitted to the lift shaft and to the aft and forward lift cranks connected to the aft/forward compression lift-rods on the forward and aft sides of the door. These will push onto the door arm to get the lift of the door. When the door is in a lifted configuration, the hold-up mechanism of the door is done through the contact of the lift roller (roller No. 2) on the lift cam in the constant radius zone of the cam. The movement of the cam at this point is irreversible and the back driving of the mechanism because of the weight of the door is stopped and kept stable by the lift cam.

Weight compensation mechanism

The function of the weight compensation mechanism is to:

- Give support to the weight of the door and the loads applied to the door during opening and closing.
- Let the door be held in the lifted position when the door is unlatched, unlocked and lifted.

The weight compensation mechanism includes the bottom hinge or door arm, stability stops and gas springs. The door arm is operated through the compression rods during the opening of the door. The stability stops keep the door in its lifted position after the door is fully unlatched, unlocked and lifted. The gas springs give support during the opening and closing and their function is to decrease the operation load during the lift and make the door operation easier.

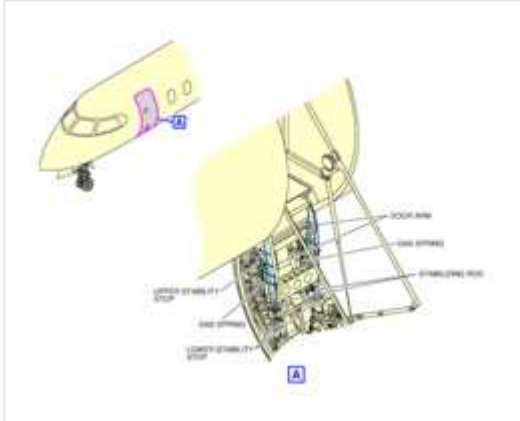


Fig 18 Weight compensation mechanism - Location

Hold-up mechanism

The hold-up mechanism has gas springs and a hold-up auxiliary device that includes the crank and cam trigger. The gas springs are attached to the hold-up auxiliary device (found on both sides of the door). The primary function of the hold-up mechanism is to:

- Make sure that the door does not fall open after the end of the closing movement and the switch is released to try to lower the door.
- Decrease the load necessary to pull the door inboard at the end of the opening movement (door lifted).

The auxiliary device holds the door to make sure the operator can lower the door in the fully closed, latched and locked position without falling open. Two rollers move into the guide fittings during the door lifting sequence. During the opening, the roller attached to the fuselage hits the cam trigger attached to the crank. This turns the cam trigger to its lower point and then activates the gas springs.

The gas springs prevent the falling of the door drop out and decrease the effect of the door drop out at the end of the lift. During the closing, the rollers attached on the fuselage side hit the cam triggers to turn them to let the rollers move into the guide fitting.

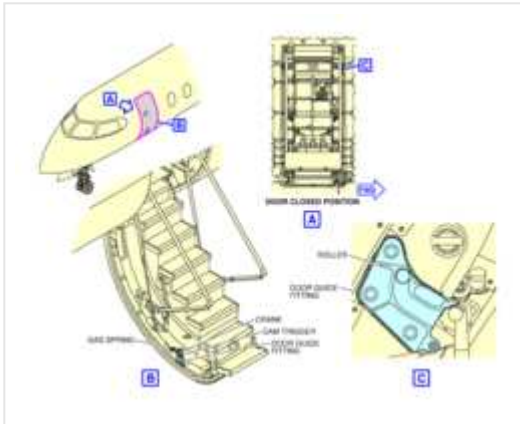


Fig 19 Hold-up mechanism - Location

Hold-open mechanism

The hold-open mechanism includes the telescopic struts, door actuator and stability stops. The function of the hold-open mechanism is to:

- Keep the door open at a safe level
- Give a damping force during the opening of the door
- Prevent the vibration of airstairs by passenger loads.

The passenger/crew door initially opens in and up. If the door is fully unlatched, unlocked and lifted, the door can be pushed outboard and will turn out smoothly around the hinge axis to the ground. The total door rotation is calculated by the length of the telescopic struts.

With the door actuator, the telescopic struts also dampen the door during the opening.

The upper stability stops blocks door movement in relation to the door arm if the door is fully opened. This keeps a stable airstair that does not rock on its hinges under the passenger weight on the airstair. The lower stability stops are commanded directly by the linkage of the handles and they are set when the door is fully lifted.

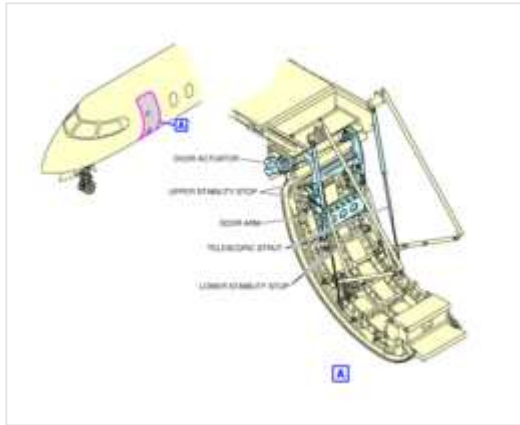


Fig 20 Hold-open mechanism - Location

Interlock system

The passenger/crew door latch-mechanism is installed with an interlock system which makes sure that the latches do not move to the latched position if the door is not in the fuselage opening.

The interlock system is made of an interlock lever installed on the internal handle shaft. The system causes the blockage of the interlock stop installed on the handrail shaft when the door is not fully closed. The interlock prevents the latching and locking functions when the door is not closed. The interlock linkage resists loads on the internal and external handles when the door is opened.

When the door is lifted and the opening has started, the interlock stop is set to cause the blockage of the interlock lever which prevents the operation of the handle mechanism with the door in the open position. During closing, the interlock stop rotates and releases the interlock lever from the blocked position and lets the handle mechanism operate.

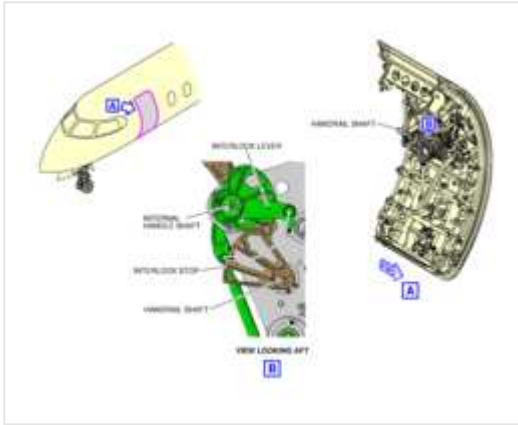


Fig 21 Interlock system - Location

System operation

The passenger/crew door operation is done through the internal and the external handles. During the operation, the door arm helps to turn the door with the help of the door actuator and the two telescopic struts.

Three proximity switches sense the closed, latched and locked position of the passenger/crew door. The aircraft electrical system supplies 28V DC power to the LGSCU. The LGSCU supplies the excitation power to the proximity switches and send the analog return signal to the DMCs. The DMCs process these signals through the internal logic for EICAS and OMS display.

Door operation

To open the door from outside the aircraft:

To open the door from outside, the external handle has to be pulled out. This opens the vent flap, unlocks, unlatches, and lifts the door. Once lifted, the door stops disengage from the fuselage and it will fall outward by gravity smoothly thanks to the actuator and the telescopic struts.

To close the door from outside the aircraft:

The door closure from outside is done by the close switch. The switch has to be pushed and held until the door reaches its full up position. Once in position, the external handle is lowered and stowed to lock and latch the door.

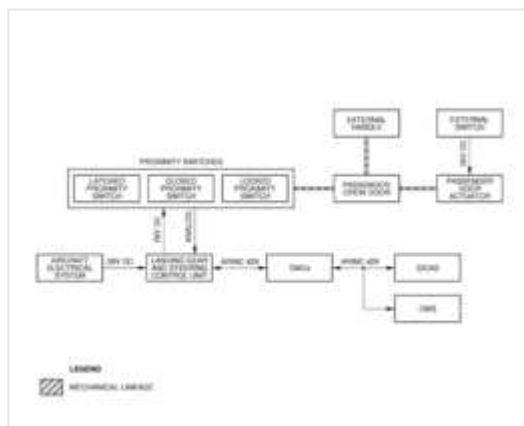


Fig 22 Passenger/crew door (external door operation) - System operation

To open the door from inside the aircraft:

Before opening the door from inside, the area below the door has to be cleared. The internal handle has to be pushed-up. This will initially open the vent flap, unlock, unlatch, and lift the door. Once lifted, the door stops disengage from the fuselage and it can be pushed-out to open. It will then fall outward by gravity smoothly thanks to the actuator and the telescopic struts.

To close the door from inside the aircraft:

To close the door from inside, the close switch has to be pushed and held until the door gets to the maximum up position. The switch is then released, the assist handle is pulled against the door surround, and the internal handle is pulled down to lock and latch the door.

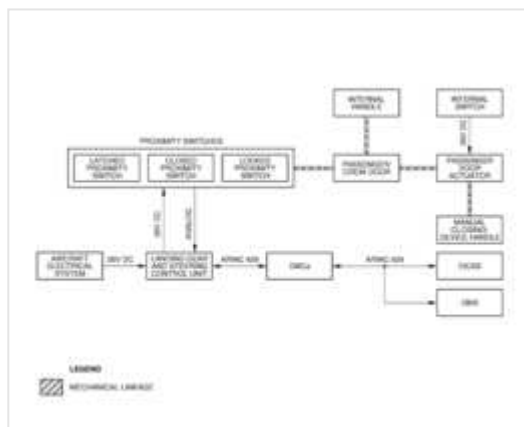


Fig 23 Passenger/crew door (internal door operation) - System operation

Manual operation

If there is an electrical problem, the door can be closed from the inside with the manual closing-device handle installed on the door actuator. To do this, the manual closing-device handle is pulled to turn a cable drum, which winds the cable and closes the door. The door is then locked with the internal or external door handle.

Control and display

The passenger/crew door is operated by the internal and external handles. The internal and external control switches operate the door actuator to close the door with electrical power. The door can be manually closed with the manual closing-device handle on the door actuator.

The door has visual indicator on the lock mechanism to show the closed and open status. The indicator is visible through the door lining. This is done independently from the proximity switches.

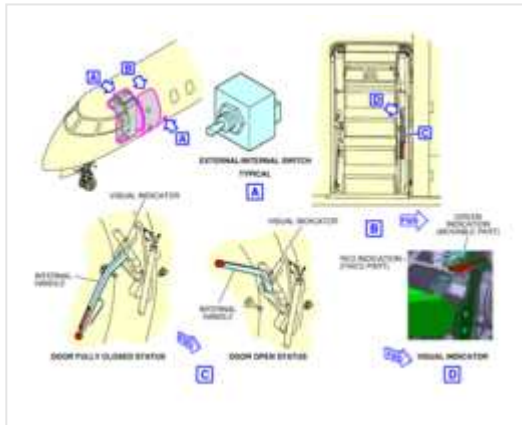


Fig 24 Passenger/crew door - Control and display

The door closed, locked, and latched status is provided by three proximity switches installed in the door mechanism. They are connected to the LGSCU channel A that send the data to the DMCs. The DMCs process the LGSCU door status in the internal logic for EICAS synoptic and Crew Alerting System (CAS) messages display.

The processed data is also sent to the OMS for status (target near, target far) and fault messages display. The status of the passenger/crew door is shown on the LGSCU-GEAR CONTROL AND DOOR STATUS DATA page of the OMS display. The status of the passenger/crew door proximity switches is shown on the LGSCU-PROXIMITY SWITCH STATUS DATA page of the OMS display.

The indications on the EICAS synoptic page are as follows:

- When the door is closed, latched and locked, the synoptic page shows a green door flush with the fuselage.
- When the door is open with no engines in operation, the door will be shown white and open.
- If there is a failure of the door or when the door is open and a minimum of one engine is in operation, the door will be shown amber and open.
- If the door is open during take-off, the door will be shown red and open.
- The indication loss of door position will be shown with a 'X' magenta icon.

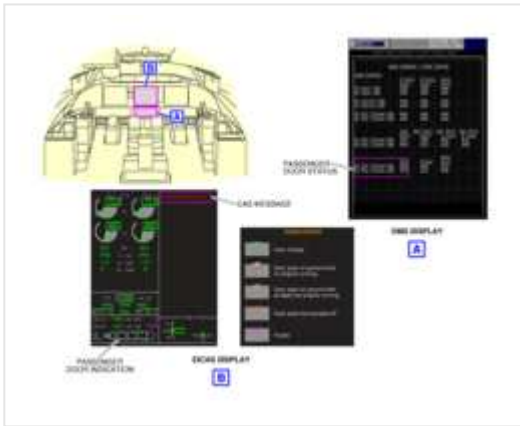


Fig 25 Passenger/crew door - EICAS and OMS displays

System interface

The passenger/crew door has interfaces with the systems and components that follow:

- The DC ESS BUS 1 (CB PNL) supplies 28V DC power to the LGSCU channel A.
- The MAIN BATT BUS supplies 28V DC power to the passenger door internal-switch.
- The passenger/crew door internal switch has electrical interfaces with the internal and external switches. This switch prevents supply of power to the closed internal and external switches when the passenger/crew door is closed, latched and locked.
- The door actuator has an electrical interface with the internal and external switches. These switches operate the door actuator to close the door from the inside or outside.
- The door actuator has a mechanical interface with the passenger/crew door to lift the door with cables and pulleys.
- The LGSCU supplies 28V DC electrical power to the proximity switches for the excitation of the proximity switches.

- The passenger/crew door proximity switches have an analog interface with the LGSCU channel A, this gives the closed, latched and locked status of the passenger/crew door to the LGSCU.
- The Integrated Air System Controllers (IASCs) of the Integrated Air Management System (IAMS) receive ARINC 429 signals by the LGSCU through DMCs about the passenger/crew door closed, latched and locked status for the temporary ramp down of the ECS flow.
- The LGSCU sends ARINC 429 signals to the DMCs for status and CAS messages display on the EICAS. Fault messages and system parameters are displayed on the OMS.
- The internal and external switches supplies 28V DC electrical power to the external lighting system installed on the passenger/crew door airstairs.
- The DMCs send the passenger/crew door status and parameters of the door proximity switches through ARINC 664 bus to the Aircraft Health Management system (AHMS) for health monitoring and maintenance.

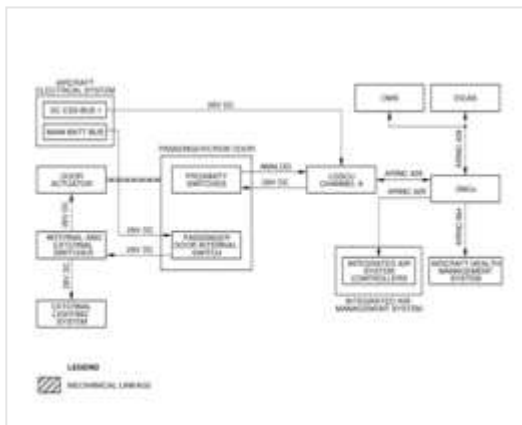


Fig 26 Passenger/crew door - System interface

For more information, refer to the Aircraft Schematic Diagram Publication (ASDP).

System monitoring

The passenger/crew door is monitored by three proximity switches to determine the door closed, door latched and door locked conditions. The LGSCU reads the status of the door proximity switches and sends the data to the DMCs, which transmits the data to the EICAS and OMS for display. The LGSCU shows that the passenger/crew door is closed, when it detects that all three proximity switches are in the active state (target near, handles in stowed position). It shows that it is open when it detects that all three proximity switches

are not in the active state (target far, handles not in stowed position). The LGSCU monitors the health status of the passenger/crew door proximity switches and sends a fault signal if the received signal is incorrect.

- There is one proximity switch for the closed condition. It is found in the door cutout and will confirm the door closed condition.
- There is one proximity switch for the latched condition. It is installed in the door closing mechanism and will confirm that the door latches are correctly attached.
- There is one proximity switch for the lock condition. It is installed in the door closing mechanism and will confirm that the door latches and mechanism are locked.

For more information about the proximity switches, refer to ATA 52-72-00 Passenger Door Warning.

The CAS and INFO messages related to the passenger/crew door shown on the EICAS are:

Table 1 - CAS and INFO messages

Message	Type	Description
PASSENGER DOOR	Warning	This indication is shown if the passenger/crew door is open during take-off.
PASSENGER DOOR	Caution	This indication is shown if the passenger/crew door is not closed, latched and locked mechanically or because of electrical failure.
DOOR OPEN	Status	This indication is shown if the passenger/crew door is open with no engine in operation.
32 GEAR-LANDING GEAR SYS REDUND LOSS	INFO	Loss of redundancy of the landing-gear control function.

05/03/21

Component location

Table 2 - Component location index

Ident	Description	Location/Zone	Maintenance	Parts	Wiring
A108	Passenger/crew door actuator	FS339L	AMP 52-11-19	IPD 52-11-19	AWM 52-11-00
PL49	Passenger/crew-door control panel	Between 361.27 and FS372.00	AMP 52-11-14	IPD 52-11-14	-
S15	Passenger/crew door external-switch	Between 361.27 and FS372.00	AMP 52-11-15	IPD 52-11-15	AWM 52-11-00
S5006	Passenger/crew door internal-switch	Zone 231 and FS305.00	SAMP 52-11-09	SIPD 52-11-09	SAWM 52-11-00

Ident	Description	Location/Zone	Maintenance	Parts	Wiring
	Passenger/crew door motor switch	Zone 830	AMP 52-11-10	SIPD 52-11-10	AWM 52-11-00
-	Passenger/crew door internal handle betalight	Zone 830	AMP 52-11-06	IPD 52-11-06	-
-	Passenger/crew door	Zone 830/Between FS320.00 and FS357.00	AMP 52-11-01	IPD 52-11-01	-
-	Passenger/crew-door key lock	Zone 830/FS339.00	AMP 52-11-03	IPD 52-11-03	-
-	Passenger/crew-door internal handle	Zone 830	AMP 52-11-05	IPD 52-11-05	-
-	Passenger/crew door telescopic-strut	Zone 830	AMP 52-11-07	IPD 52-11-17	-
-	Passenger/crew door external-handle	Zone 830/FS339.00	AMP 52-11-13	IPD 52-11-13	-
-	Passenger/crew door handrails	Zone 830	AMP 52-11-17	IPD 52-11-17	-
-	Passenger/crew door airstairs	Zone 830	AMP 52-11-21	IPD 52-11-21	-
-	Passenger/crew-door motor cable	Zone 830	AMP 52-11-23	IPD 52-11-19	-
-	Passenger/crew door seal	Zone 830	AMP 52-11-25	IPD 52-11-25	-
-	Passenger/crew door gas-spring	Zone 830	AMP 52-11-27	IPD 52-11-27	-
-	Passenger/crew door top-step	Zone 830	AMP 52-11-33	IPD 52-11-33	-
-	Passenger/crew door bottom-step	Zone 830	AMP 52-11-35	IPD 52-11-35	-
-	Passenger/crew-door vent-flap seal	Zone 830	AMP 52-11-37	IPD 52-11-37	-
-	Passenger/crew door side-covers	Zone 830	AMP 52-11-39	IPD 52-11-39	-
-	Passenger/crew door vent-flap	Zone 830	AMP 52-11-41	IPD 52-11-41	-
-	Airstair upper side-cover	Zone 830	AMP 52-11-43	IPD 52-11-39	-
-	Airstair mid side-cover	Zone 830	AMP 52-11-45	IPD 52-11-39	-
-	Airstair lower side-cover	Zone 830	AMP 52-11-47	IPD 52-11-39	-
-	Passenger/crew-door handrail-pulley cover	Zone 830	AMP 52-11-49	IPD 52-11-49	-
-	Passenger/crew-door area step-trim-panel	Zone 830	SAMP 52-11-53	SIPD 52-11-53	-
-	Passenger/crew-door storage bins	Zone 830	AMP 52-11-57	IPD 52-11-57	-
-	Passenger/crew door handrail extension	Zone 830	AMP 52-11-18	IPD 52-11-18	-
-	Passenger/crew door roller pin (lower)	Zone 830	AMP 52-11-59	IPD 52-11-59	-

Ident	Description	Location/Zone	Maintenance	Parts	Wiring
-	Passenger/crew door roller pin (upper)	Zone 830	AMP 52-11-61	IPD 52-11-61	-